

Frame Dominance Measurement Framework

A Comprehensive Guide to the Four-Component Method

Conference Presentation Materials

1 Overview: The Multi-Method Dominance Framework

Our framework identifies **dominant frames** in media discourse through a comprehensive four-component approach. Each component contributes 25% to the final dominance score, ensuring no single dimension can artificially inflate a frame's importance.

1.1 Core Principle

Dominant frames must demonstrate multiple forms of centrality:

1. **Information Structure:** They carry substantial information that shapes interpretation
2. **Network Position:** They occupy central positions as conceptual hubs
3. **Causal Influence:** They drive the appearance of other frames
4. **Sustained Presence:** They maintain consistent proportional presence

1.2 Consensus Requirement

A frame achieves dominant status only when selected by **at least 3 out of 4 methods**. This consensus approach ensures robustness and prevents methodological artifacts.

2 Component 1: Information Theory Analysis (25%)

2.1 What It Measures

Information theory analysis evaluates how much each frame contributes to the overall information structure of the discourse. It answers: *"How much does knowing about this frame tell us about the entire discourse?"*

2.2 Three Key Metrics

2.2.1 Mutual Information

Definition: Quantifies the average information shared between one frame and all others.

Mathematical Formula:

$$MI(X, Y) = \sum_{x, y} p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right)$$

Interpretation:

- High mutual information means the frame’s presence strongly predicts other frames
- Example: If the political frame appears, we gain significant information about whether economic or scientific frames will also appear
- Frames with high MI are ”information hubs” that structure the discourse

Implementation Details:

- Frame proportions are discretized using adaptive binning: $n_{bins} = \min(10, \max(3, \lfloor \sqrt{N} \rfloor))$
- Computed for each frame pair, then averaged across all pairs
- Captures both positive and negative associations

2.2.2 Entropy Reduction

Definition: Measures how much knowing a frame’s state reduces uncertainty about the overall system.

Mathematical Formula:

$$ER = \frac{H(X) - H(X|Y)}{H(X)}$$

where $H(X)$ is joint entropy and $H(X|Y)$ is conditional entropy.

Interpretation:

- High entropy reduction indicates the frame provides substantial information about discourse structure
- A frame that always co-occurs with specific configurations has high entropy reduction
- Think of it as ”predictive power” - how well does this frame predict the overall discourse state?

2.2.3 Information Content

Definition: Self-entropy measuring temporal variability and unpredictability.

Mathematical Formula:

$$H(X) = - \sum_i p(x_i) \log p(x_i)$$

Interpretation:

- High information content = high variability over time
- Frames that appear and disappear carry more information than constant frames
- Balances against frames that are simply ”always there” without structure

2.3 Conference Presentation Tip

Use the analogy of a conversation: Some words (frames) are like ”therefore” or ”however” - when they appear, they tell us a lot about what’s coming next. These are high information frames.

3 Component 2: Network Analysis (25%)

3.1 What It Measures

Network analysis identifies structurally central frames by treating discourse as a complex network where frames are nodes and co-occurrences are edges.

3.2 Network Construction

- **Time Windows:** 52-week sliding windows with 50% overlap
- **Edge Creation:** Frames co-occurring above 0.1 proportion threshold
- **Edge Weights:** Joint probability $w_{ij} = P(i > 0.1 \cap j > 0.1)$

3.3 Four Network Metrics

3.3.1 PageRank Centrality

Definition: Google's PageRank algorithm applied to frame networks.

Mathematical Formula:

$$PR(i) = \frac{1-d}{N} + d \sum_{j \in M(i)} \frac{PR(j)}{L(j)}$$

where $d = 0.85$ (damping factor), N = number of nodes, $M(i)$ = frames linking to i , $L(j)$ = outbound links from j .

Interpretation:

- Frames are important if connected to other important frames
- Captures "influence through association"
- Political frame scores high because it connects to all major frames

Why It Matters: Identifies conceptual hubs that organize discourse structure.

3.3.2 Eigenvector Centrality

Definition: Measures connectivity to highly-connected frames.

Mathematical Principle: A frame's centrality is proportional to the sum of centralities of frames it's connected to.

Interpretation:

- Like being "popular with popular people"
- Complements PageRank by emphasizing quality of connections
- Identifies frames in the "inner circle" of discourse

3.3.3 Temporal Stability

Definition: Inverse coefficient of variation of centrality measures over time.

Formula:

$$Stability = \frac{1}{CV} = \frac{\mu}{\sigma}$$

Interpretation:

- High stability = consistent network position
- Distinguishes truly dominant frames from temporarily central ones
- Frames that maintain centrality across decades score higher

3.4 Conference Presentation Tip

Use social network analogy: Some people (frames) are influential because everyone knows them (PageRank) and they know other influential people (Eigenvector), and they maintain this position over time (Stability).

4 Component 3: Causality Analysis (25%)

4.1 What It Measures

Identifies frames that drive the temporal evolution of other frames - the "agenda setters" of discourse.

4.2 Two Complementary Approaches

4.2.1 Granger Causality Testing

Definition: Statistical test for whether past values of frame X help predict future values of frame Y.

Method:

1. Fit Vector Autoregression (VAR) model
2. Test if lagged values of X improve prediction of Y
3. Use F-statistic to assess significance

Adaptive Lag Selection:

- $N < 10$: 1 lag
- $N < 20$: 2 lags
- $N < 40$: 3 lags
- $N \geq 40$: 4 lags
- Optimal lag chosen by Akaike Information Criterion (AIC)

Metrics Computed:

- **Breadth:** Proportion of frames that X Granger-causes
- **Strength:** Average F-statistic across significant relationships

Interpretation: High Granger causality means the frame "leads the conversation" - when it appears, others follow.

4.2.2 Transfer Entropy

Definition: Information-theoretic measure of directional information flow.

Mathematical Formula:

$$T_{X \rightarrow Y} = \sum p(y_{t+1}, y_t^k, x_t^l) \log \frac{p(y_{t+1} | y_t^k, x_t^l)}{p(y_{t+1} | y_t^k)}$$

Key Advantages:

- Captures non-linear relationships Granger might miss
- No assumption of linear dynamics
- Measures actual information transfer

Interpretation:

- High transfer entropy = frame exports information to discourse
- Identifies "trigger frames" that initiate cascades
- Political frame shows high TE, suggesting agenda-setting role

4.3 Why Both Methods?

- Granger captures linear temporal dependencies
- Transfer entropy captures complex, non-linear patterns
- Together provide comprehensive causality picture

4.4 Conference Presentation Tip

Think of it as identifying who starts trends: Some frames are followers (low causality), others are leaders (high causality) that set the agenda for discourse.

5 Component 4: Proportional Dominance (25%)

5.1 What It Measures

Sustained quantitative presence in discourse - not just frequency, but quality of presence.

5.2 Four Complementary Metrics

5.2.1 Mean Proportion

Definition: Average presence across entire time period.

Formula:

$$\bar{p}_i = \frac{1}{T} \sum_{t=1}^T p_{i,t}$$

Interpretation:

- Baseline measure of overall prevalence
- Political frame: 35% average presence
- Economic frame: 28% average presence

5.2.2 Elevation Score

Definition: How much frame exceeds median of all frames.

Formula:

$$Elevation = \max\left(0, \frac{p_i - m}{m}\right)$$

where p_i = frame's mean proportion, m = median across all frames.

Interpretation:

- Identifies frames consistently above baseline
- Normalized measure allowing comparison
- Penalizes frames hovering around median

5.2.3 Consistency Score

Definition: Inverse coefficient of variation rewarding stable presence.

Formula:

$$Consistency = \frac{1}{1 + \sigma_i/\mu_i}$$

Interpretation:

- High consistency = reliable discourse component
- Low consistency = episodic, event-driven appearance
- Distinguishes sustained dominance from spiky presence

5.2.4 Percentile Rank

Definition: Position in overall distribution of mean proportions.

Interpretation:

- Robust to outliers
- Provides normalized 0-100 scale
- Easy to interpret: 90th percentile = top 10%

5.3 Why All Four?

Each metric captures different aspect:

- Mean = raw presence
- Elevation = above-average performance
- Consistency = reliability
- Percentile = relative position

Together they identify frames with substantial, elevated, and consistent presence.

5.4 Conference Presentation Tip

Like measuring influence in a meeting: Not just who talks most (mean), but who consistently (consistency) speaks more than average (elevation) and ranks among top speakers (percentile).

6 Integration: The Consensus Approach

6.1 Computing Final Dominance

6.1.1 Step 1: Normalize Each Component

Each of the four components produces scores normalized to $[0,1]$ range.

6.1.2 Step 2: Equal Weighting

$$Composite_Score = 0.25 \times Info + 0.25 \times Network + 0.25 \times Causality + 0.25 \times Proportional$$

6.1.3 Step 3: Apply Four Statistical Methods

1. **Above Median:** Frames scoring above median composite score
2. **Natural Break:** Maximum gradient in sorted scores (elbow method)
3. **Cumulative Variance:** Point explaining 70% of variance
4. **Statistical Outliers:** Mean + 0.5 standard deviations

6.1.4 Step 4: Consensus Requirement

Frame is dominant if selected by ≥ 3 of 4 methods.

6.2 Why This Approach?

Robustness:

- No single method can determine dominance
- Reduces sensitivity to methodological choices
- Increases confidence in results

Theoretical Grounding:

- Dominance is multifaceted phenomenon
- Requires evidence across multiple dimensions
- Reflects complexity of discourse dynamics

7 Practical Application: Results

7.1 Overall Period (1978-2024)

Frame	Info	Network	Causality	Proportional	Status
Political	1.00	0.73	1.00	1.00	Dominant
Economic	0.59	0.71	0.46	0.72	Dominant
Scientific	0.71	0.35	0.84	0.47	Near-miss
Environmental	0.43	0.30	0.50	0.27	Not dominant

7.2 Key Findings

- **Political frame:** Dominant across all dimensions
- **Economic frame:** Strong network and proportional presence
- **Scientific frame:** High information and causality, but declining network centrality

8 Conference Presentation Strategy

8.1 Opening Statement

"We developed a comprehensive framework that identifies dominant frames through four independent lenses - information theory, network analysis, causality testing, and proportional dominance. A frame must excel in at least three to be considered truly dominant."

8.2 Key Talking Points

8.2.1 On Methodology

- "Each method captures different aspect of dominance"
- "Consensus approach ensures robustness"
- "Equal weighting prevents methodological bias"

8.2.2 On Results

- "Political frame emerges as unequivocally dominant"
- "Economic frame shows strong co-dominance"
- "Scientific frame's network marginalization is concerning"

8.2.3 On Implications

- "Methodology exportable to other policy domains"
- "Provides empirical grounding for paradigm theory"
- "Reveals how discourse structures shape policy debates"

8.3 Handling Questions

Q: Why equal weighting?

A: "We chose equal weighting to avoid imposing theoretical priors. Each dimension captures fundamentally different aspect of dominance. Sensitivity analysis shows results robust to reasonable weight variations."

Q: Why 3/4 consensus threshold?

A: "Majority consensus (3/4) balances stringency with flexibility. Requiring all 4 would be too restrictive; requiring only 2 would be too lenient. This threshold identifies frames with broad-based dominance."

Q: How does this relate to Hall's paradigm theory?

A: "We operationalize Hall's abstract concept by identifying the specific configuration of dominant frames that constitute the paradigm. Our multi-method approach captures the institutionalized patterns Hall describes."

Q: What about other frames not shown as dominant?

A: "Non-dominant frames still play important roles but don't structure the overall discourse. They appear conditionally, often triggered by specific events or in conjunction with dominant frames."

9 Technical Appendix: Implementation Details

9.1 Data Requirements

- Minimum 52 weeks of data for annual analysis
- Frame proportions at sentence level
- Temporal resolution: weekly aggregation

9.2 Computational Considerations

- Information theory: $O(n^2)$ for pairwise MI
- Network analysis: $O(n^3)$ for eigenvector centrality
- Causality: $O(n^2p)$ where p = lag order
- Total runtime: 5 minutes for 47 years of data

9.3 Software Implementation

- Python 3.8+ with NumPy, SciPy, NetworkX
- Statsmodels for Granger causality
- PyInform for transfer entropy

- Parallel processing for efficiency

10 Conclusion

This four-component framework provides the first systematic empirical approach to measuring frame dominance in media discourse. By combining information theory, network analysis, causality testing, and proportional dominance metrics, we can identify which frames truly structure discourse versus those that merely appear frequently.

The consensus requirement ensures robust results while the equal weighting prevents any single methodological perspective from dominating. This approach is:

- **Theoretically grounded:** Each component maps to theoretical concept of dominance
- **Empirically rigorous:** Multiple validation through different methods
- **Practically applicable:** Exportable to other domains and contexts
- **Reproducible:** Clear metrics and implementation details

For conference presentation, emphasize that this framework bridges the gap between abstract paradigm theory and empirical measurement, providing scholars with tools to track how dominant interpretive structures evolve over time.